MILITARY SPECIFICATION

CONTROL, RECEIVER-TRANSMITTER C-9887( )/GRA-114

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Control, Receiver-Transmitter C-9887( )/GRA-114 which provides remote operation of Receiver/Transmitter RT-1185, part of Radio Data Link, Sound Ranging AN/GRA-114 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Materials-handling (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock Container Grade and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-45	Tape, Gummed Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD- TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

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MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

CC-A-678035	Burn-in Cycle
SM-A-865232	Performance Test Procedures

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 (see 6.2(g)).

3.2 Construction. The Control, Receiver-Transmitter C-9887( ) /GRA-114 (Remote Unit) shall be constructed in accordance with the requirements of this specification and the model.

3.2.1 Printed circuit boards. Where feasible, printed circuit boards shall be provided with pull-out flanges. Flanges shall be labeled.

3.2.1.1 Test points. Test points shall be provided on the printed circuit board. Test points shall be properly labeled and located so as to facilitate testing without removal of the board from the case.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Power switch. Power switch shall be provided with a power indicator to show when it is in the ON condition.

3.4 Finish. All surfaces of the C-9887 requiring a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel markings shall be group I as described in that specification.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each C-9887 shall be serial-numbered (see 4.4).

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. The C-9887 shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678035 (see 4.7).

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3.8 Performance requirements.

3.8.1. Bias, standby. The voltage shall be 1.0V ± 10 percent when the line bias supply is 2.4 Vdc, the remote power supply is 18.0 Vdc, and the push-to-talk switch is in the receive position (see 4.9).

3.8.2 Voice bias and gain. The D.C. bias voltage shall be between 0.50 to 3.2 VDC negative. The amplifier shall provide an 0.8 V rms ±20% output for a 1.0 mv input signal. Gain shall remain constant down to 12 VDC power supply voltage and with the remote leads reversed (see 4.9).

3.8.3 Code bias and duration. The duration of the signal shall be 150 to 500 ms. The polarity of the signal across the line terminals shall be positive and no less than 4.0 volts at any battery supply voltage down to 12 volts.

3.8.4 Voice distortion. The distortion shall not be greater than 2.5 percent (see 4.9).

3.8.5 Voice response. The frequency response shall comply with Table I (see 4.9).

TABLE I. Voice response.

Frequency (Hz)	Response (dB)
150	-3±1
300	-1±1
600	0±1
1000	0 Reference
3500	-.5±1
5000	-1.5±1

3.8.6 Line receiver distortion and gain. The distortion shall not be greater than 2.5 percent at supply voltages of 12-18 V (see 4.9).

3.8.7 Frequency response, line receiver. The frequency response shall comply with Table II. (see 4.9)

TABLE II. Frequency response.

Frequency (Hz)	Response (dB)
150	-3±1
300	-1±1
600	0±.5
1000	0 Reference
3500	-.5±1
5000	-1.5±1.5

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3.9 Service conditions. The C-9887 shall meet the service conditions below.

3.9.1 Temperature.

3.9.1.1 High temperature. During testing as specified in 4.10.1.1, the C-9887 shall meet the following requirements:

Paragraph	Performance
3.8.1	Standby bias
3.8.2	Voice bias and gain
3.8.3	Code bias and duration
3.8.4	Voice distortion
3.8.5	Voice response
3.8.6	Line receiver distortion and gain
3.8.7	Frequency response, line receiver

3.9.1.2 Low temperature. During testing as specified in 4.10.1.2, the C-9887 shall meet the following requirements:

Paragraph	Performance
3.8.1	Standby bias
3.8.2	Voice bias and gain
3.8.3	Code bias and duration
3.8.4	Voice distortion
3.8.5	Voice response
3.8.6	Line receiver distortion and gain
3.8.7	Frequency response, line receiver

3.9.2 Humidity. (See 4.10.2) The C-9887 shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials and shall meet the following performance requirements:

Paragraph	Performance
3.8.1	Standby bias
3.8.2	Voice bias and gain
3.8.3	Code bias and duration
3.8.4	Voice distortion
3.8.5	Voice response
3.8.6	Line receiver distortion and gain
3.8.7	Frequency response, line receiver

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3.9.3 Immersion. The C-9887 shall show no evidence of leakage after it is tested as specified in 4.10.3.

3.9.4 Dust. After testing as specified in 4.10.4 all controls and fittings shall operate smoothly and the C-9887 shall meet the following performance requirements:

Paragraph	Requirements
3.8.4	Voice distortion
3.8.6	Line receiver distortion and gain

3.9.5 Vibration, bounce and shock.

3.9.5.1 Bounce and shock. The C-9887 shall meet the requirements of Table III.

TABLE III. Bounce and shock.

Paragraph Heading	Test Para	Performance After Test
Bounce	4.10.5.1	Specified performance. 1/ No physical damage except minor surface abrasion.
Shock, drop	4.10.5.2	Specified performance 1/. Any physical damage shall be minor only.

1/ The receiver-transmitter control shall meet specified performance for the following:

Paragraph	Requirement
3.8.3	Code bias and duration
3.8.4	Voice distortion
3.8.6	Line receiver distortion and gain

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3.9.6 Salt fog. After the salt fog test of 4.10.6, the C-9887 visually, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface. Electrical operation is not required.

3.9.7 Fungus. After the fungus test of 4.10.7, the C-9887 shall be visually examined and shall reveal no evidence of viable fungus on the surface.

3.10 Weight. The C-9887 shall weigh no more than 2.75 pounds, less batteries (see 4.11).

3.11 Electromagnetic interference characteristics. The C-9887 shall meet the emission and susceptibility requirements for MIL-STD-461 for Class I equipments. Test set-ups and test procedures shall conform to the measurement techniques of MIL-STD-462 (see 4.12). The following requirements are applicable:

Transmit mode	Receive mode	
CE06	CS03	CE06
RE02 1/	CS04	RE02 1/
RS03 1/	CS05	RS03 1/

1/ The upper frequency of the test shall be 1000 Mhz.

3.12 Visual and Mechanical. The C-9887 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.4):

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning of units
General requirements for securing of parts
Cleaning prior to welding
Controls
Self tapping screws
Cleaning after welding
Printed wiring assemblies

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.8).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article unit. The contractor shall furnish five (5) first article units of the Control, Receiver-Transmitter C-9887( )/GRA-114.

4.3.2 First article inspection. The first article inspection shall consist of the inspection specified in Table IV, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE IV. First article inspection.

Inspections	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
1. Inspections covered by subsidiary documents 1/	3.3,3.4 3.5,3.12	3.4	Inspection to be performed on all units				
2. Burn-in 1/	3.7	4.7	Inspection to be performed on all units				
3. Group A inspection 1/	See Table V		Inspection to be performed on all units				
4. Group B inspection 1/	See Table VI		Inspection to be performed on all units				
5. Group C inspection 2/							
High temperature	3.9.1.1	4.10.1.1	2				
Low temperature	3.9.1.2	4.10.1.2	1				
Humidity	3.9.2	4.10.2	3				
Immersion	3.9.3	4.10.3			2		
Dust	3.9.4	4.10.4				1	
Bounce	3.9.5.1	4.10.5.1				2	
Shock, drop	3.9.5.1	4.10.5.2			3		
Salt fog	3.9.6	4.10.6		2			
Fungus	3.9.7	4.10.7					1
Weight	3.10	4.11		1			
EMI	3.11	4.12			1		

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns in the table, covered by inspection 5, specify the order of inspections for the indicated unit.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract or purchase order.

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Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.12

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table V. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table V.

4.5.1.1 Order of inspection within Group A. A Group A inspection shall be performed in an order satisfactory to the Government.

TABLE V. Group A inspection.

Inspection	Reqt Para	Test Para	AQL Major	AQL Minor
Visual and mechanical	3.12	4.13	1%	4%
Electrical				1/
Bias, standby	3.8.1	4.9	1%	
Voice bias and gain	3.8.2	4.9	1%	
Code bias and duration	3.8.3	4.9	1%	
Voice distortion	3.8.4	4.9	1%	
Line receiver distortion and gain	3.8.6	4.9	1%	

1/ All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VI and to the special inspection levels of Table 1 of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sample plans, for the AQLs listed in Table VI, shall be as follows:

AQL	Inspection level
4%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE VI. Group B inspection.

Inspection	Requirement Paragraph	Test Paragraph	AQL
Interchangeability	3.6	4.6	6.5%
Voice response	3.8.5	4.9	4%
Frequency response, line receiver	3.8.7	4.9	4%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Five (5) samples of the C-9887 shall be selected at random, one (1) sample for each subgroup in Table VII from the first fifty (50) units or fraction thereof produced. The fungus tests shall be performed only on the first production lot. Thereafter, four (4) samples of C-9887 shall be selected at random from every fifty (50) units or fraction thereof greater than or equal to 25 units, one (1) sample for each subgroup 1 through 4 of Table VII. If the fraction is less than 25, the selection of Group C samples for the preceding test shall include units from the remaining fraction.

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TABLE VII. Group C inspection.

Inspection	Requirement Paragraph	Test Paragraph
Subgroup 1		
High temperature	3.9.1.1	4.10.1.1
Low temperature	3.9.1.2	4.10.1.2
Humidity	3.9.2	4.10.2
Subgroup 2		
Dust	3.9.4	4.10.4
Bounce	3.9.5.1	4.10.5.1
Subgroup 3		
EMI	3.11	4.12
Immersion	3.9.3	4.10.3
Shock; drop	3.9.5.1	4.10.5.2
Subgroup 4		
Salt fog 1/	3.9.6	4.10.6
Subgroup 5		
Fungus 1/	3.9.7	4.10.7

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.3.2 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection, in that order.

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4.6 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Burn-in. Each C-9887 submitted for acceptance shall be subjected to burn-in in accordance with drawing SM-A-678035 (see 3.7).

4.8 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.9 Performance characteristics. The performance characteristics of Control, Receiver-Transmitter C-9887( )/GRA-114 shall be tested in accordance with the Performance Test Procedure, Drawing SM-A-865232. Paragraph titles in the test procedure correspond to the requirement paragraphs of the specification.

Requirement	Paragraph
Bias, standby	3.8.1
Voice bias and gain	3.8.2
Code bias and duration	3.8.3
Voice distortion	3.8.4
Voice response	3.8.5
Line receiver distortion and gain	3.8.6
Frequency response, line receiver	3.8.7

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4.10 Service condition. The C-9887 shall be tested in accordance with the following conditions:

4.10.1 Temperature.

4.10.1.1. High temperature. The C-9887 shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. For Step 4, the chamber temperature shall be 71°C (160°F). For Step 7, the chamber temperature shall be adjusted to 65.6°C (150°F). (This includes effects of solar radiation) (see 3.9.1.1).

4.10.1.2 Low temperature. The C-9887 shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -45.6°C (-50°F), and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be -40°C (-40°F) (see 3.9.1.2).

4.10.2 Humidity. The C-9887 shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 7 of Procedure III, the chassis shall be removed from its enclosure, or the test item shall be otherwise opened so as to expose to the chamber environment the normally sealed areas. In Step 6 measurements shall be taken during the last five hours of each test cycle. In Step 8 measurements shall be taken the last 5 hours of each 48 hours (see 3.9.2).

4.10.3 Immersion. The C-9887 shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 (see 3.9.3).

4.10.4 Dust. The C-9887 shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810 (see 3.9.4).

4.10.5 Vibration, bounce and shock.

4.10.5.1 Bounce. The C-9887 shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810. It shall undergo one hour of testing at each of the following temperatures: (1) Room ambient; (2) -45°C, (-50°F); (3) 71°C, (160°F).

At each of these temperatures it shall be vibrated for ten minutes on each face (see 3.9.5.1).

4.10.5.2 Shock, drop. The C-9887 equipment shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810 (see 3.9.5.1).

4.10.6 Salt fog. The C-9887 shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810 (see 3.9.6).

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4.10.7 Fungus. The C-9887 shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. Gasket sealed or nonhermetic sealed test items shall be opened during exposure and internal areas, as well as external surfaces, shall be sprayed with spore suspension. Operation of the equipment is not required (see 3.9.7).

4.11 Weight. The C-9887 shall be weighed and shall meet the requirement of 3.10.

4.12 Electromagnetic interference characteristics. The C-9887 shall be tested in accordance with MIL-STD-462 and meet requirements of 3.11.

4.13 Visual and mechanical inspection. The C-9887 shall be examined for the applicable defects listed in MIL-STD-252.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.(2)(b)).

5.1.1 Level A.

5.1.1.1 Cleaning Each Control, Receiver-Transmitter C-9887( )/GRA-114 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each control, Receiver-Transmitter C-9887( )/GRA-114 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Control, Receiver-Transmitter C-9887( )/GRA-114. Each control shall be individually packed method III as follows: cushion the unit by wrapping in paperboard conforming to PPP-P-291, type 1, style 1. Secure the cushioning with tape conforming to PPP-T-45, type III, class 1, grade A. Wrap the technical literature, packaged as specified in 5.1.1.4.1, around the unit and place the unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservation application and unit packaging. Each control shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

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5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.(2)(b)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Control, Receiver-Transmitter C-9887 ( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping, conforming to QQ-S-781, type I, finish A or shrink film.

5.2.2 Level B. A quantity of Control, Receiver-Transmitter C-9887 ( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers shall be pallet-mounted on pallets conforming to NN-P-71,

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type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2.1 Unitization. Palletization shall be required when: containers specified in 5.2.2 do not require pallet mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.2, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The Control, Receiver-Transmitter C-9887( )/GRA-114 (Control Unit) provides a method for the remote control of Receiver-Transmitter RT-1185/GRA-114 (Sound Observer) component of Radio Data Link AN/GRA-114. The operator maintains control and communication through keying button and handset.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see Section 5).

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c. When rough handling test is required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design, for example, C-9887W/GRA-114. The contractor should apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

Custodian:

Army-EL

Preparing Activity:
Army-EL
Project No. 1290-A334

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-C-49114(EL) CONTROL, RECEIVER-TRANSMITTER, C-9887( )/GRA-114

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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